Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech | *Semester: VI* | *Time: 03 Hours*
Code: 100613 | *Subject: Electrical Drives* | *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Write the short answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) What is an electrical drive?
(b) List out the drawback of rectifier fed DC drives.
(c) What is steady state stability for an electric drive?
(d) What is regenerative braking?
(e) Write the speed control methods of DC drives.
(f) What is Slip recovery scheme in induction motor?
(g) Define PWM.
(h) Write the Starting torque equation of three phase induction motor.
(i) What is the most economical method of electric braking?
(j) Why the thyristor control is preferred over Ward Leonard system of speed control?

Q.2 A 220 V, 800 rpm, 60A DC separately excited motor has an armature resistance of 0.09 Ω. It is broken by plugging from an initial speed of 1500 rpm. Calculate (i) Resistance to be placed in armature circuit to limit braking current to twice the full load value (ii) Braking torque (iii) Torque when the speed has fallen to zero. **[14]**

Q.3 (a) Explain reverse voltage braking of a three phase induction motor drive; also give the speed-torque curve for it. **[7]**
(b) What are the disadvantages of DC drives due to which the three phase induction motor is replacing it? **[7]**

Q.4 What is slip power Recovery scheme? Explain the operation with suitable diagram and derive the expression of motor torque. **[14]**

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Q.5 (a) With schematics explain different types of self excited generators. **[7]**
(b) What are the modes of operation in electric drive system? **[7]**

Q.6 (a) Explain three phase controlled converters with neat sketch? **[7]**
(b) Explain briefly about flux weakening operation of induction motor drives? **[7]**

Q.7 What do you understand by constant torque drive and constant power drive? Explain the various methods of braking of induction motors. **[14]**

Q.8 Draw the block diagram of an electric drive. Explain the function of power modulator in detail. **[14]**

Q.9 Write Short notes on ***any two*** of the following: **[7 x 2= 14]**
(a) Loss calculation in dc motor
(b) Operating regions of dc machine
(c) Torque-speed curve of pump load

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